



CASEL In Higher Education

AI Guided Socratic Dialogue for Developing Social Emotional Learning (SEL) Skills

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174
Students

9
Institutions

4
Groups

5
CASEL domains

- Social Emotional Learning (SEL)
- CASEL Framework
- AI Guided Socratic Dialogue
- CASELy Platform
- Randomized Controlled Trial

The CASEL Framework for SEL
Social Emotional Learning (SEL) is the process of acquiring and applying skills that support emotional regulation, positive social interaction, intrinsic motivation, resilience, and responsible decision-making.



The Problem:
An Urgent Need for SEL in Higher Education

Escalating Challenges
University students face increasing emotional and psychological challenges

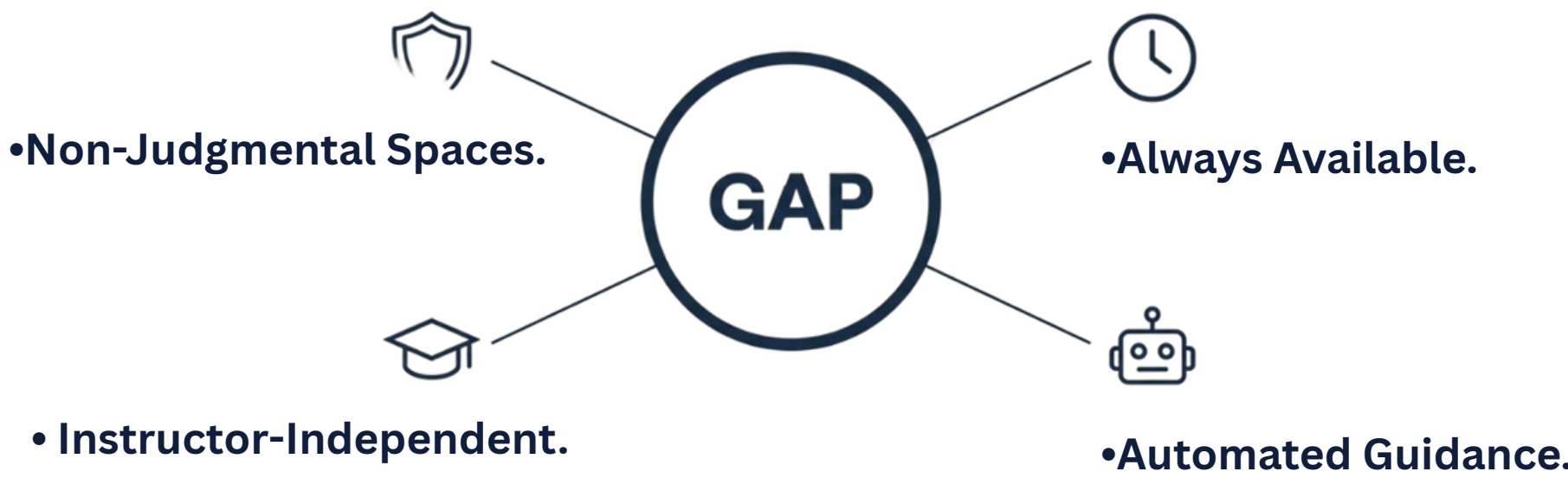
Impact on Academic Success
These difficulties threaten well-being and create significant barriers to academic engagement and performance.

Limitations of Existing Support
Many students lack practical, accessible, non-judgmental SEL practice opportunities.

RESEARCH QUESTION

How does an AI-based Socratic dialogue influence the development of SEL skills among higher education students?

Current SEL Tools in Higher Education:
The Gap



There are no suitable tools that combine these features for higher education students.

What Is a Socratic Bot?

Definition
•Inquiry through open-ended questions
•Promotes reflection and self-awareness

The Socratic Bot in Our Study
•AI-based conversational agent
•Supports self-reflection
•No advice, instruction, or evaluation

Methodology

174
Participants · 9 institution

101
Advanced students (multi-institution)

73
First year students (Braude, Course 251961)

RCT
Randomized Controlled Trial

15 March 2026
Ethics Approval

Recruitment Context

Advanced Students
Voluntarily recruited from nine different academic institutions.

First Year Students
Recruited through the Engineering Fundamental Skills course (Course 251961) at Braude College.

Randomized Controlled Trial (RCT)

3 Experimental Groups
• Group A (Experimental)
• Group B (Experimental)
• Group C (Experimental)

1 Control Group
• Group D (Control)

Participants were randomly assigned (1:1:1:1) to the four groups.

Participant Demographics

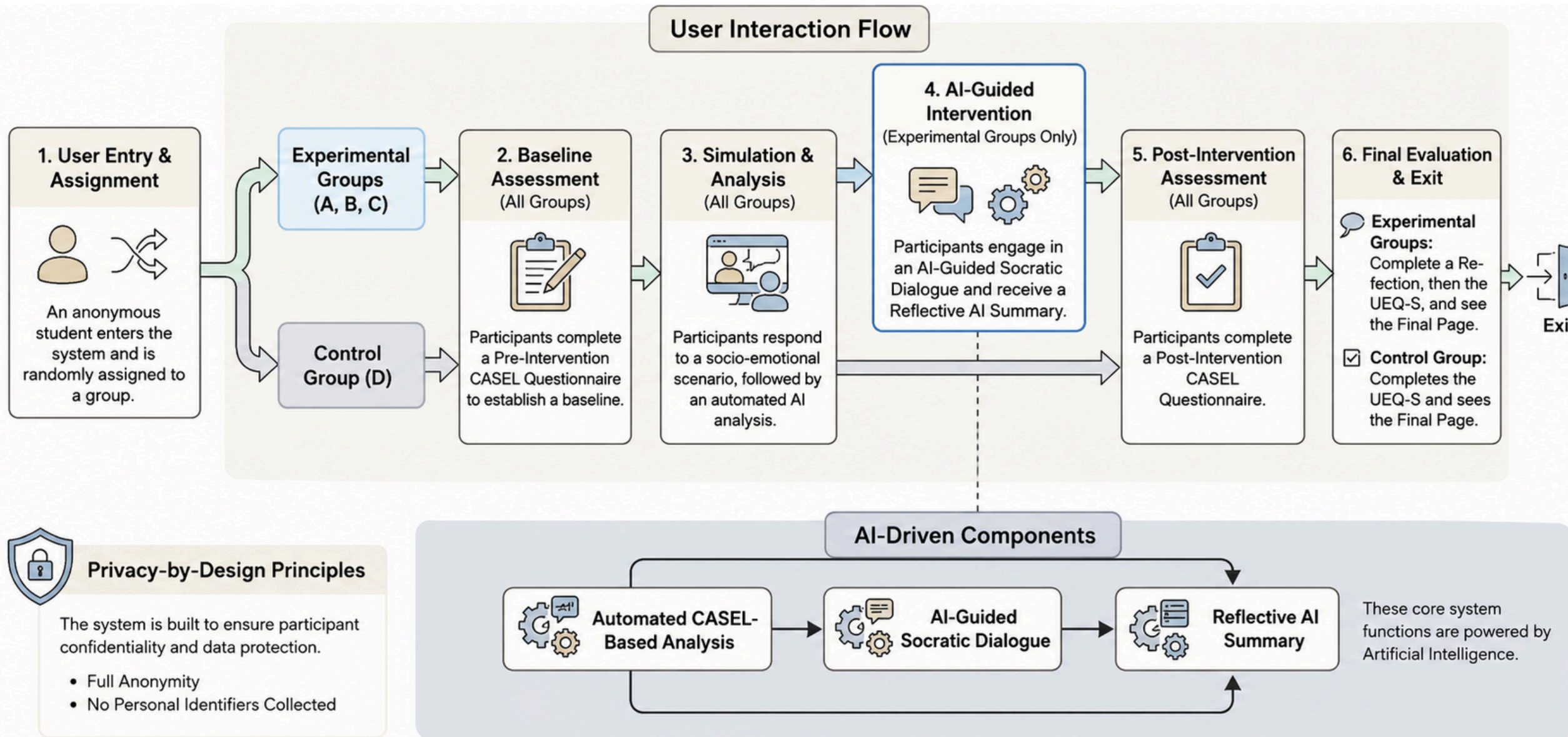
Gender Distribution (N=174)
88 Male 50.6%
85 Female 48.9%
1 Other / Not Specified (0.6%)

Institutional Distribution (N=174)
146 Braude College 83.9%
28 Other Institutions 16.1%

Age Distribution (N=174)
18-22 (41.4%)
23-26 (43.7%)
27-30 (12.1%)
31-35 (2.3%)
36+ (0.6%)

Research Group Assignment (N=174)
Group A (Experimental) 45 (25.9%)
Group B (Experimental) 43 (24.7%)
Group C (Experimental) 43 (24.7%)
Group D (Control) 43 (24.7%)

Experimental Procedure



Experimental Procedure Highlights

Linear & Closed Architecture
Sequential workflow with no skipped stages.

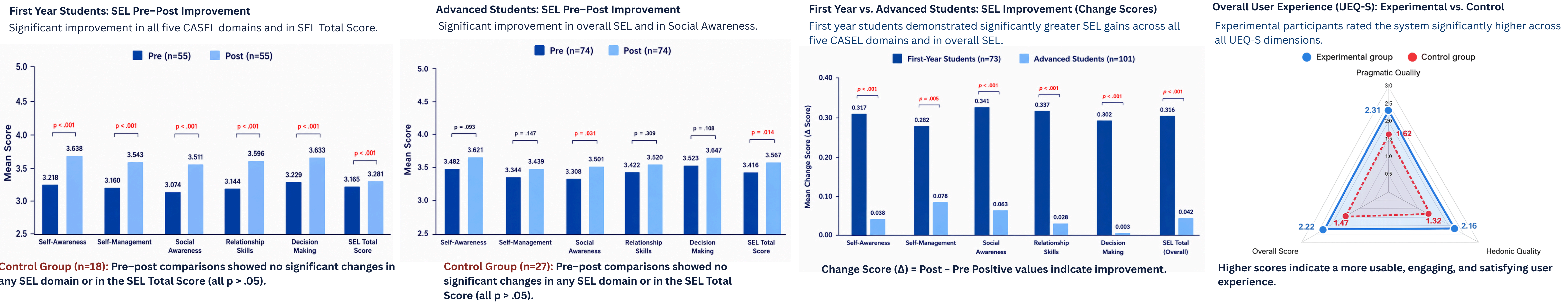
1:1:1:1 Randomization
Automatic allocation to four study groups.

Privacy by Design
Anonymous UUIDs with no personal identifiers.

Validated Simulation
Academic socio-emotional scenario activating all five CASEL domains.

Results

Quantitative Results



Qualitative Results

Key Thematic Findings:
The "Pause" Technique
Fact vs. Emotion
Redefining Help Seeking
Integrity Under Pressure

Cohort Differences
First Year Students
Advanced Students

Demo
1. Automatic Reaction (Stress / Fear)
2. The Socratic "Pause" (Reflection)
3. Value- Aligned Choice (Responsibility)